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Role: Software Development Engineer in Test

Online Test

Question 1:

Get Maximum Memory Difference (Greedy)

You are given an array of integers and an integer k, you have to select k elements from the array so that the sum of absolute difference between each pairs in the k selected elements is the maximum. You must return the maximum sum.

Example:

Input: array = [2,3,1,2] k = 3

Output: 4

Explanation:

$$[2,1,2] = |2 - 1| + |1 - 2| + |2 - 2| = 2$$

$$[2,3,1] = |2 - 3| + |2 - 1| + |3 - 1| = 4$$

$$[2,3,2] = |2 - 2| + |2 - 3| + |3 - 2| = 2$$

$$[3,1,2] = |3 - 1| + |1 - 2| + |3 - 2| = 4$$

The answer is 4.

Question 2:

Minimal String

Given a string "s", you have to use two other strings "pre" and "post" to make "post" the most lexicographically minimal string and return "post". But you can only perform two operations :

1. Remove the first element of "s", and append it to "pre".
2. Remove the last character from pre and append it to "post"

Example:

Input: s = "acdb" Output: "abdc"

Explanation:

s	pre	post
"acdb"	""	""
"cdb"	"a"	""
"cdb"	""	"a"
"db"	"c"	"a"
"b"	"cd"	"a"
""	"cdb"	"a"
""	"cd"	"ab"

""	"c"	"abd"
""	""	"abdc"

The answer is post = "abdc"

17 candidates were shortlisted for the Technical Interview 1.

Technical Interview 1

1. Give an algorithm to detect cycle in a linked list.
2. Remove duplicate elements from a sorted linked list.
[Remove Duplicates from Sorted List - LeetCode](#)
3. Delete all nodes that have duplicate numbers, leaving only distinct numbers from the original list.
[Remove Duplicates from Sorted List II - LeetCode](#)
4. Find the square root of a number.
[Sqrt\(x\) - LeetCode](#)
5. Given a string containing zeros and ones give a substring consisting of k ones. The length of the substring must be minimal and it should be lexicographically smaller.

And some basic questions from DSA, DBMS, OS and CA

1. What is Kernel? Which part of the OS is Kernel?
2. What is virtual memory?
3. What is polymorphism?

10 candidates were shortlisted for the HR Interview.

HR Interview

General question like:

1. Tell me about yourself
2. How can you contribute to software testing?
3. Given a developer role and a testing role, what would you choose?
4. Where do you see yourself in 5 years?
5. Tell me the technical skills in which you are strong and weak.
6. Given a testing role in our company and developer role in some other company for the same salary what would you choose?

7 candidates were short listed for the Technical Interview 2

Technical Interview 2

Question 1:

Design a database for the home page of an e-commerce website considering all possible cases. Also give the data structures to be used.

Question 2:

You are asked to build an application to overwrite or append multiple files. What approach would you use ensure the atomicity property (either everything happens or nothing happens)?

Question 3:

Given an array arr with pattern $arr[0] < arr[1] < \dots < arr[i-1] < arr[i] > arr[i+1] > \dots > arr[n]$ find the index i.